COMP 4221 NLP Course Project Report

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Project Scope

This project implements the full noisy-news workflow required by the assignment: light preprocessing for an unlabeled corpus, retrieval with two classical bag-of-words ranking models, and a small supervised topic-classification experiment built on a directly reviewed label set that was later refined through a consensus-assisted external LLM audit workflow.

Task 1: Corpus Audit and Preprocessing

- Documents: `871` - Total cleaned tokens: `1,147,881` - Average document length: `1317.89` - Median document length: `1250.00` - Vocabulary size before preprocessing: `57,401` - Vocabulary size after preprocessing: `47,998`

The corpus parser splits documents with the leading `@@<numeric_id>` marker. Cleaning is intentionally light: repeated gap noise is mapped to `<GAP>`, text is lowercased, and whitespace is normalized while punctuation, stopwords, and rare tokens are preserved for downstream interpretability.

Task 2: Retrieval

- Average TF-IDF `P@5`: `0.76` - Average BM25 `P@5`: `0.78` - TF-IDF wins: `3` - BM25 wins: `3` - Ties: `4` - Best BM25 advantage: `Which articles report on basketball, boxing, golf, baseball, or other sports competitions and athlete performance?` (`0.80` vs `0.40`) - Best TF-IDF advantage: `Which articles cover movies, television, theater, books, or entertainment industry figures?` (`0.80` vs `0.60`)

The retrieval comparison shows a narrow BM25 advantage overall, but the result is query-dependent rather than absolute. BM25 performs better on compact, topic-heavy queries, while TF-IDF remains stronger when exact lexical overlap dominates the information need.

Task 2 Qrels Validation

- Validation passed: `True` - Covered ranked rows: `100/100` - Missing qrels rows: `0` - Explicit label mismatches: `0`

Task 3: Topic Classification

- Labeled documents: `200` - Labeling method: `single_reviewer_curated_with_external_llm_audit` from `direct_review` - Review summary path: `/Users/nagato/workspace/project/outputs/task3_label_review_summary.json`
- Review tag: `direct_review_v2` - Corrected labels after direct review: `19`

- Additional changes from the previous reviewed release: `13` - Baseline test accuracy / macro-F1: `0.8667` / `0.8561` - Improved test accuracy / macro-F1: `0.9333` / `0.9330` - Selected SVM `C`: `1.0`

Label Distribution

- Politics: 43 - Culture/Entertainment: 37 - Sports: 32 - Business/Economy: 32 - Crime/Law: 31 - Other: 25

External LLM Audit

- Providers: `vertex, xai` - Shared audited documents: `200` - Cross-provider agreement rate: `0.905` - Unanimous disagreements against the released labels: `0`

Main Error Pattern

- Crime/Law -> Other (1 cases): `5022441` - Politics -> Crime/Law (1 cases): `4114241`

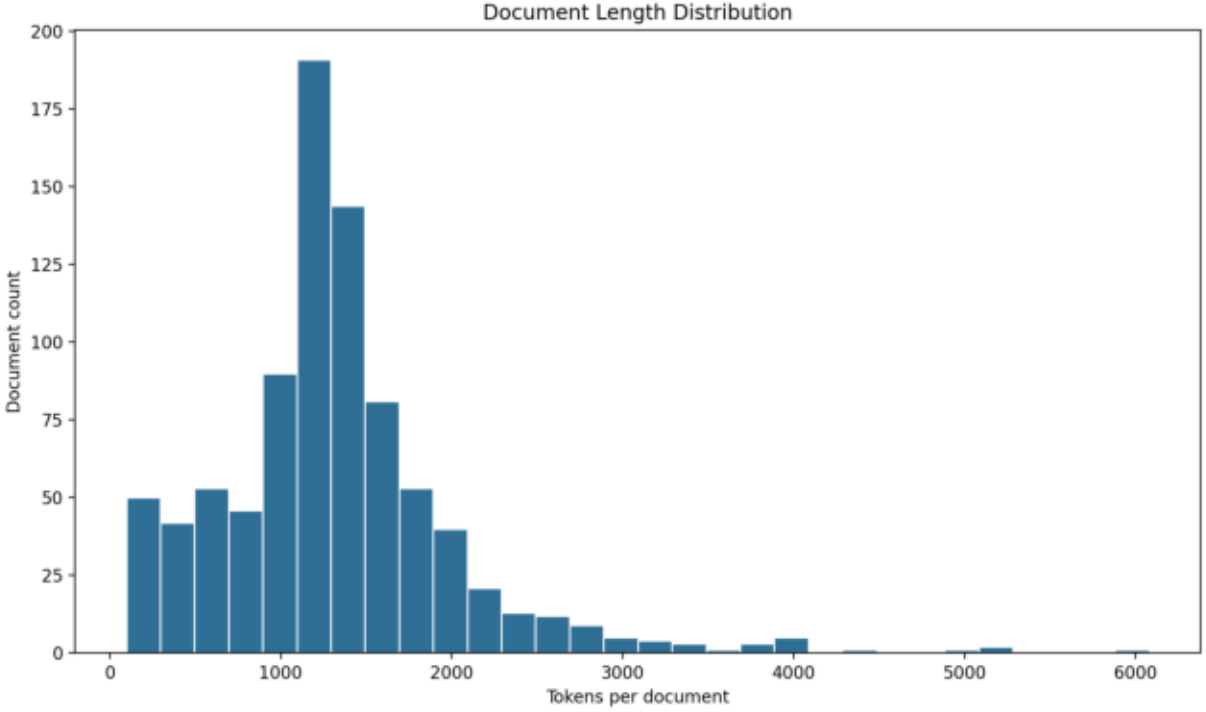
The classification experiment now uses a directly reviewed label file that was then re-adjudicated on a subset of externally flagged disagreements. It is therefore stronger than the earlier direct-review-only release, though it still falls short of adjudicated multi-human gold labels.

Reproducibility Notes

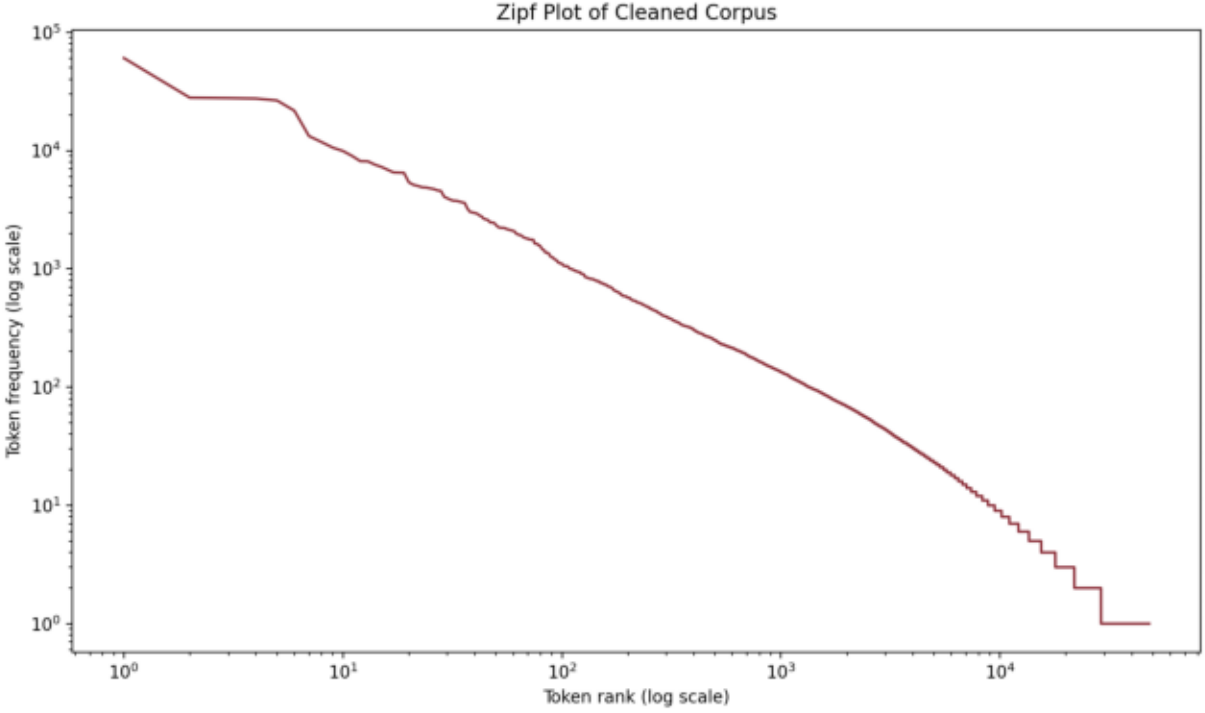
- The current Task 3 labels were directly reviewed after document inspection and then re-adjudicated on unanimous external-audit disagreements, and the correction summary is stored in `outputs/task3_label_review_summary.json`. - The external audit itself is preserved in `outputs/task3_external_reviews/` and summarized in `outputs/task3_llm_audit_summary.json`. - The earlier bootstrap recipe remains available in `configs/task3_bootstrap.yaml` as an auxiliary seed-generation artifact, but it is no longer the authoritative label source. - Retrieval judgments are stored in a model-independent qrels file keyed by query-document pairs. - The repository can be rebuilt with shell scripts rather than notebooks, and all main entry points expose explicit CLI arguments.

Figures

Doc Length Hist



Zipf Plot


Zipf Plot of Cleaned Corpus
Token frequency (log scale)
Token rank (log scale)
10^0
10^1
10^2
10^3
10^4
10^5

Confusion Matrix

